

Autonomous First Responder Fire & Rescue Robot with Virtual Payload Delivery and Acoustic Localization

CSE 396 Computer Engineering Project - Project Proposal Report

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Group: 7

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1 Introduction

1.1 Problem

In high-risk fire and disaster environments, it is critically dangerous for human rescue teams to conduct initial area reconnaissance. While existing rescue robots can navigate these areas, physical payload delivery significantly increases the robot's hardware cost, power consumption, mechanical failure rate, and overall weight.

Furthermore, relying on cloud-based AI for scene analysis is impossible in disaster zones where internet infrastructure is destroyed. Therefore, the specific problem this project solves is **developing an autonomous, Edge-AI-driven "First Responder" reconnaissance robot**. This robot enters the disaster zone before firefighters to map the area and continuously locates victims utilizing Tri-Modal Sensor Fusion (Acoustics, Vision, and Environmental Data). It classifies victim severity (trapped, lying down, standing) and provides initial support (two-way audio, wake-up protocols, augmented status reports, and SOS beaconing) utilizing strict on-device computing (Edge AI) and a zero-risk Dual Powerbank Architecture.

2 Literature Review

Traditional search and rescue robots heavily rely on teleoperation and deterministic navigation. Recent advancements in Edge AI have introduced Vision-Language Models (VLMs), object detection models (YOLO), and lightweight Speech-to-Text (STT) models directly onto single-board computers, enabling semantic understanding without internet access.

For victim localization, Acoustic Source Localization (ASL) using microphone arrays is a proven method for finding conscious victims. Initially, this project considered utilizing a Thermal Sensor Matrix to combat the Tyndall effect in dense smoke where RGB cameras fail. However, due to prohibitive hardware costs and the inherent safety risks of replicating real fire/smoke conditions for physical testing, thermal imaging was deprecated.

Instead, this project addresses the localization challenge by fusing **Acoustic Homing** with an **Adaptive Edge-AI Pipeline**. To ensure real-time performance on edge constraints, the system will empirically compare the efficacy of a quantized VLM against a custom-trained Convolutional Neural Network (CNN) and a rule-based YOLO-Pose engine, selecting the optimal architecture based on latency and accuracy.

3 Methods

3.1 Data

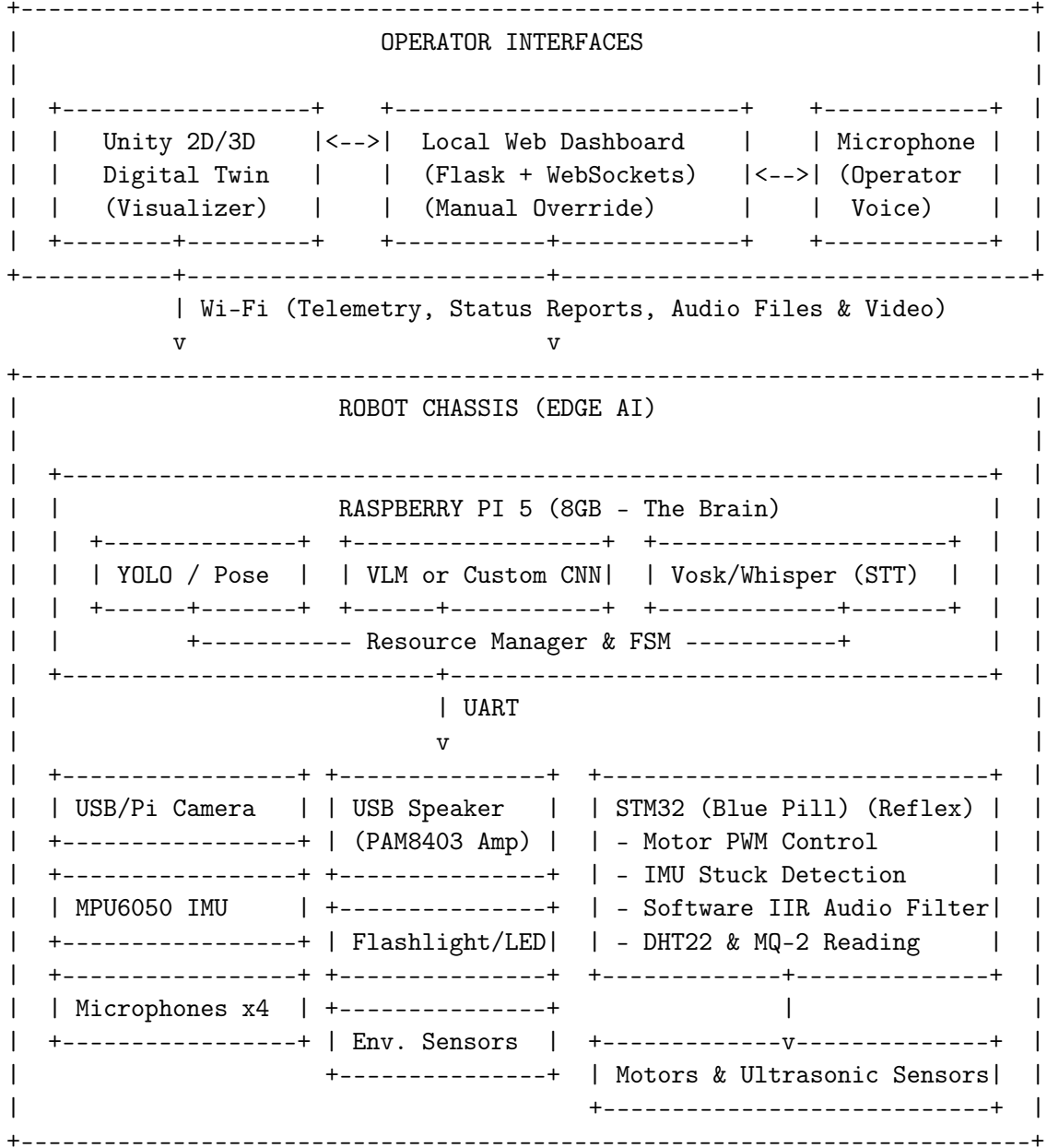
The system will strictly operate on real-time sensory data acquired from the environment; no external APIs will be queried.

- **Visual Data:** Live RGB frames captured via a USB/Pi Camera. Used for target detection and severity analysis.
- **Continuous Acoustic Data:** Analog audio signals continuously monitored by an array of directional microphones (MAX4466), sampled via the STM32 ADC.
- **Voice Command Data:** .wav audio files recorded by the operator on the Web Dashboard for local STT processing.
- **Environmental & Spatial Data:** Real-time metrics from MQ-2 (Smoke), DHT22 (Temperature), MPU6050 IMU (Movement/Stuck Detection), and HC-SR04 (Ultrasonic).

3.2 Methodology

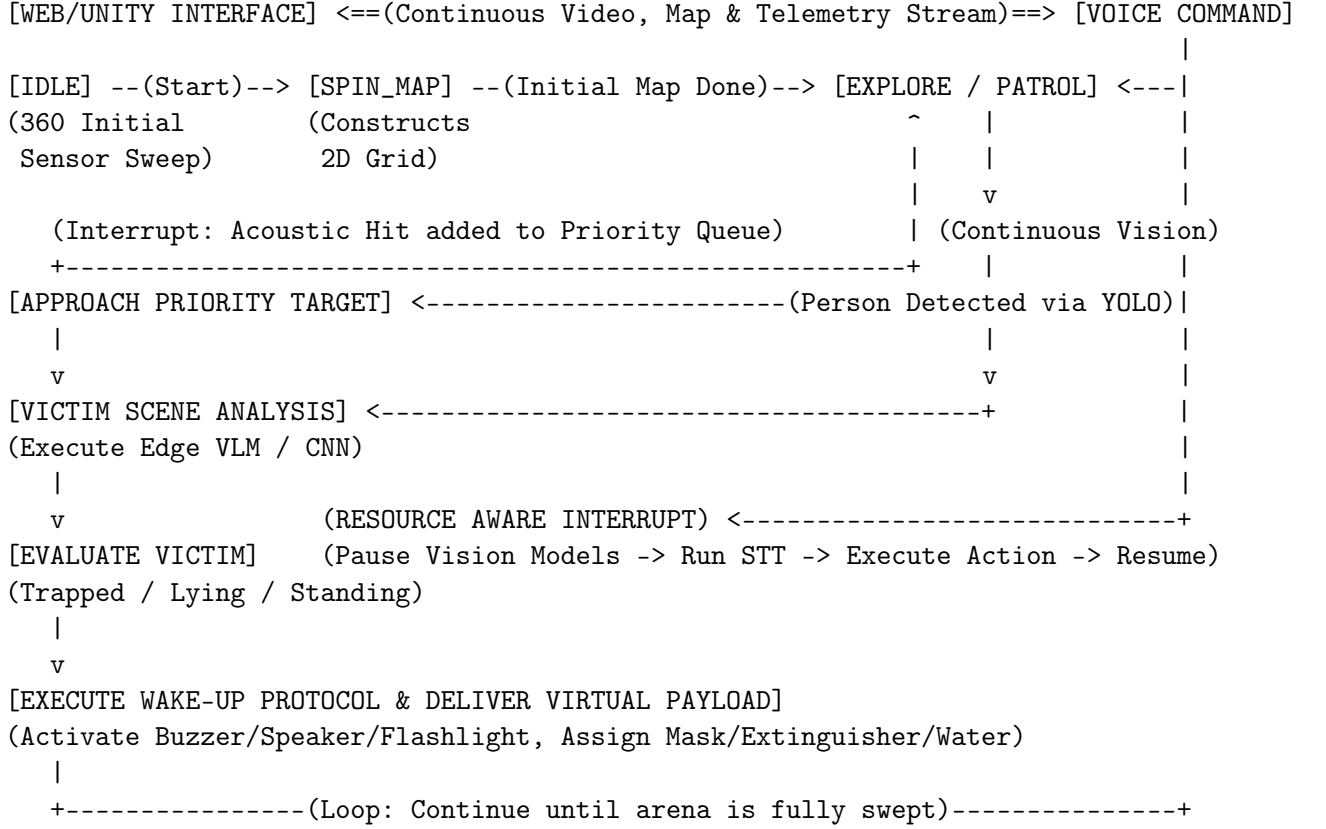
3.2.1 System Architecture Overview

The project utilizes a **Hybrid Edge Architecture**. A Raspberry Pi 5 handles all AI and networking, while an STM32 manages real-time hardware reflexes. To eliminate the fire risks associated with Li-Po batteries in disaster robotics, the system is powered by a **Dual PD Powerbank Setup**, utilizing a 12V Type-C Decoy trigger for the mobility chassis.



3.2.2 Tri-Modal Search & Dynamic Priority Flow

The baseline behavior of the robot is to explore the **entire** arena. Target detection triggers the Finite State Machine (FSM) via three modalities: Vision (YOLO), Acoustic (Microphones), or Environmental (Sensors). Detected locations are added to a **Priority Queue**, temporarily hijacking the exploration path to assess the victim.



3.2.3 Dynamic Resource Management & Failsafes

To prevent the Pi 5 from overheating and to guarantee safe operations, we implement core optimizations:

- **Dynamic FPS Management:** During standard "Explore" mode, the camera operates at a low framerate (e.g., 5 FPS). Upon detecting a trigger, the FSM boosts computing power to 100% for victim assessment.
- **Time-Based Return-to-Home (RTH):** Because modern PD Powerbanks regulate their output to a constant voltage until total depletion, traditional ADC voltage-reading failsafes are ineffective. Therefore, the FSM implements a strict time-based RTH protocol (e.g., automatically returning to the starting point after 60 minutes of operation) to prevent stranding.
- **Dead Man's Switch:** If the WebSocket connection drops, the FSM triggers an immediate RTH, reversing the last 30 seconds of telemetry to regain the Wi-Fi signal.
- **Power-Saving Beacon Mode:** Once a trapped/unconscious victim is secured, the robot powers down heavy AI pipelines and motors, sustaining only the SOS light beacon and two-way audio.

3.2.4 Target Prioritization & Wake-up Protocol

Once a target is approached, the AI pipeline evaluates the frame. If multiple people are found, the FSM sorts targets in a Priority Queue:

1. **Trapped / Lying Down (High Priority):** The robot approaches immediately. It executes the **Wake-Up Protocol** (flashing SOS lights, buzzer alarm, and two-way operator speaker) to assess consciousness. It locks the map location with a **RED/YELLOW pin** on Unity.

2. **Standing (Lower Priority):** The robot evaluates the environment (Smoke/Heat) and assigns a virtual payload (e.g., Smoke Mask, Extinguisher), marking the map with a **GREEN** pin.
3. **Proximity Tie-Breaker:** If multiple targets of the *same* priority level exist in a single frame, the system navigates to the *closest* target first (estimated via YOLO bounding box area or ultrasonic depth).
4. **Augmented Status Report:** The robot transmits a comprehensive JSON report containing: Exact coordinates, MQ-2 Oxygen/Smoke estimate, DHT22 Temperature, and Victim Status.

3.2.5 Techniques

- **Software-Based Digital IIR Filtering:** To cancel continuous motor noise from the acoustic data, we evaluated hardware (RC) filters versus software digital filters. We selected a Software-based Digital IIR (Infinite Impulse Response) Filter executed on the STM32. Unlike static hardware filters, the IIR filter dynamically attenuates unwanted frequencies with minimal computational overhead (simple multiplications and additions) after ADC sampling.
- **Edge STT Processing & Preemptive Scheduling:** Utilizing offline models (Vosk/Whisper.cpp) via a preemptive resource scheduling algorithm. When a voice command is received, heavy vision models are strictly **paused** to prevent Out-Of-Memory errors, prioritizing the STT pipeline.
- **Tri-Modal Sensor Fusion:** Fusing Visual cues (YOLO), Acoustic Bearings (Mic Arrays), and Spatial/Environmental constraints (Ultrasonic + IMU + DHT22) to guarantee navigation.
- **Adaptive Edge Vision Pipeline:** Comparing three methodologies to find the optimal Edge AI balance: a quantized VLM (Moondream2), a Custom Transfer-Learned CNN, and a rule-based YOLO-Pose engine.
- **Systematic Exploration (Spin-Scan):** An initial 360-degree rotation (SPIN_MAP) to populate a preliminary 2D occupancy grid using ultrasonic sensors, followed by a frontier-based sweeping algorithm.

3.3 Hardware Requirements (BOM)

Component	Model/Description	Purpose
Brain (SBC)	Raspberry Pi 5 (8GB) + Active Cooler	Edge AI Pipeline, Web Server, FSM.
Storage	64GB Micro-SD Card (A2 Class)	High IOPS storage for OS, Swap Memory, and AI Models.
Reflex (MCU)	STM32F103C8T6 (Blue Pill)	Motor control, IIR Audio Filtering, ADC reading.
Vision	Pi Camera Module V3	Frame capturing for AI.
Acoustics	3x MAX4466 Mics & PAM8403 Amp + Speaker	Sound localization & Operator two-way audio.
Sensors	HC-SR04, MQ-2, DHT22, MPU6050 (IMU)	Navigation, Smoke, Temp, and Stuck Detection.
Movement	4WD Chassis + L298N Driver	Mobility platform.
Power (Pi 5)	10000mAh PD Powerbank (5V/3A)	Stable, risk-free power dedicated to the Pi 5.
Power (Motors)	10000mAh PD Powerbank + 12V Type-C Decoy	Regulated 12V output for DC motors, eliminating Li-Po fire hazards.

3.4 Module Structure and Task Distribution (8-Person Team)

Person	Mod 1: Emb & HW	Mod 2: AI & Vision	Mod 3: Audio & Nav	Mod 4: Web & STT	Mod 5: Unity
Gabil	Secondary	Primary			
Ziya	Primary				Secondary
Dicle		Secondary	Secondary		Primary
Ömer	Secondary			Primary	
Uğur		Secondary	Primary	Secondary	
Tuana		Secondary	Secondary	Primary	
Evrin		Primary	Secondary		Secondary
Fatma		Primary		Secondary	

3.4.1 Module Responsibilities

- **MODULE 1: Embedded & Hardware (2 People):** Physical assembly and real-time hardware reflexes via the STM32 microcontroller. Driving DC motors using PWM, polling environmental sensors, and reading the MPU6050 IMU for stuck detection. Managing the Dual Powerbank (12V Decoy) distribution and UART communication.
- **MODULE 2: AI & Vision Pipeline (2 People):** Managing the visual perception and classification layer on the Raspberry Pi 5. Integrating the Pi Camera and running continuous YOLO inference with dynamic FPS management. Executing the Edge VLM (or Custom CNN) for victim severity assessment. Implementing the pause/resume logic during STT interrupts.
- **MODULE 3: Acoustics & Navigation (2 People):** Sound processing, directional hearing, and autonomous movement logic. Implementing the Software-based Digital IIR Filter on the STM32 to calculate distress call bearings. Developing the initial Spin-Scan mapping algorithm, AND-logic obstacle avoidance, and the Acoustic Homing protocol.
- **MODULE 4: Web Dashboard & STT (1-2 People):** Operator control interface, communication bridging, and Voice Command processing. Developing the Flask/WebSockets

backend to stream live video and telemetry. Integrating the offline STT model (Vosk/Whisper.cpp). Structuring the Augmented Status Reports (JSON).

- **MODULE 5: Unity Digital Twin (1-2 People):** 3D visualization and situational awareness for the operator. Building the Unity C# dashboard to reflect the robot's real-time position. Implementing the Priority Queue mapping system to drop color-coded pins based on victim status, and designing the Push-to-Talk UI.

4 Evaluation & Deliverables

4.1 Evaluation

1. **AI Latency & Model Selection:** The chosen vision architecture (VLM vs. CNN vs. YOLO-Pose) must accurately classify target severity and resolve in under 5 seconds entirely on the edge.
2. **Acoustic Reliability:** The Digital IIR Filter must attenuate motor noise effectively enough to detect a human distress call.
3. **Failsafe Executions:** The IMU stuck detection, Time-Based RTH, and Wi-Fi RTH protocols must trigger correctly under adverse conditions.
4. **Digital Twin Synchronization:** Unity must reflect the augmented status reports and color-coded priority pins with minimal latency.

4.2 Deliverables (Quarterly Expectations)

Must Have (Minimum Viable Product):

- A functional physical robot driven by STM32, utilizing AND-Logic obstacle avoidance.
- Raspberry Pi 5 successfully running an offline Vision AI (YOLO + Classifier) to detect and assess humans.
- A Web Dashboard streaming the camera feed and rudimentary telemetry.

Should Have (Target Success):

- Successful deployment of the Augmented Status Report and Wake-Up Protocols (Buzzer + Speaker) for Trapped/Lying victims.
- Dynamic FPS management and Resource-Aware STT pausing for optimal RAM/Thermal performance.
- Acoustic Source Localization (SSL) with Digital IIR Filtering to redirect the robot towards sound.
- Unity 2D/3D Digital Twin tracking the robot's movement and dropping color-coded pins.

Nice to Have (Bonus / Optional):

- Full integration of the Time-Based RTH and Dead Man's Switch protocols.
- Power-Saving Beacon Mode for extended battery life during rescue waits.

4.3 Time Plan & Syllabus Milestones

Note: Hardware procurement has been shifted to overlap with initial software architecture development to ensure continuous progress without hardware bottlenecks.

Week	Phase / Milestone	Deliverable
W 1-3	Project Ideation & Structuring	Concept finalized, team modules assigned, BOM created.
W 4 (15/3)	Project Proposals	Document Submission. Hardware orders placed.
W 5	PC-Based Dev & Hardware Setup	Parts arrive. Pi OS / STM32 setup. Unity UI design and AI testing start.
W 6 (24/3)	Module Requirements (Headers)	API contracts finalized. C/C++ .h files and Python interface structures pushed to Git.
W 7	Sub-system Integration	STM32 reading sensors and driving motors. Pi testing Digital IIR Filter.
W 8 (7/4)	Module documentation	Technical documentation of algorithms and wiring diagrams complete.
W 9	Acoustic & Failsafe Testing	Robot prioritizes targets (handling proximity tie-breakers), executes wake-up protocols, and handles RTH/STT interrupts.
W 10 (21/4)	Module demonstrations	Individual unit tests presented to the instructor.
W 11-13	Full Integration & UI Polish	WebSockets fully integrated; physical arena end-to-end testing and edge case debugging.
W 14 (19/5)	Final Project Demonstration	Video presentation, user manual, and website. The last grade will be mainly based on the final demonstration on week 14.

4.4 Risk Analysis and Mitigation

Risk	Prob/Impact	Mitigation Strategy
Pi 5 Overheating / OOM	High / High	Use Active Cooler. Dynamic FPS throttling. Pausing Vision models during STT inference.
Acoustic Noise / Echo	High / Med	Implement Software-based Digital IIR Filter in STM32 to attenuate motor frequencies.
Stranding (Power Loss)	Low / High	Utilize Dual PD Powerbanks with a Time-Based Return-to-Home (RTH) protocol (e.g., 60-minute limit).
VLM Too Slow	High / High	Evaluate and fallback to Custom CNN or YOLO-Pose rule-based engine.

5 References

- [1] Instructor: Salih Sarp
- [2] CSE 396 Computer Engineering Project Syllabus, Gebze Technical University.
- [3] Ultralytics. (2023). YOLOv8 Documentation for Edge Devices.
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- [5] Valin, J. M., et al. (2007). Robust Sound Source Localization Using a Microphone Array on a Mobile Robot. *IEEE/RSJ International Conference on Intelligent Robots and Systems*.
- [6] Unity Technologies. (2023). Unity Networking and UI Toolkit Documentation.